

# Blacktip Refine — AWS Marketplace Setup Guide

A complete, end-to-end walkthrough for deploying **Blacktip Refine** from the AWS Marketplace into **your own AWS account**. Refine is self-hosted: the software runs on one EC2 instance in your environment, your cost and inventory data never leave your account, and there is no SaaS to send data to.

**Time:** ~20-30 minutes, mostly an ~8-minute CloudFormation create. **Cost:** Refine is an **annual contract subscription on AWS Marketplace**, billed through your AWS account. Pick the tier that covers your total annual cloud spend ("Spend Analyzed") — Starter \$7,500 ( $\leq \$100K$ ), Core \$20,000 ( $\leq \$500K$ ), Growth \$50,000 ( $\leq \$1M$ ), or Scale \$125,000 ( $\leq \$5M$ ); larger estates use an Enterprise Private Offer. On top of the subscription you pay only for the small EC2 instance (~\$30-60/mo on `t3.medium`).

**Editions:** **Free** — \$0 for up to **\$1,000/month** of analyzed cloud spend, self-hosted, funded by an anonymous aggregated monthly telemetry summary (total spend + savings only); it pauses with an upgrade prompt over the cap. **Paid** — the annual contract tiers above (by Spend Analyzed), strictly telemetry-free.

## What gets deployed

The CloudFormation template provisions, in **your** account:

- **1 EC2 instance** (default `t3.medium`) running the Refine container, pulled from AWS Public ECR.
- **1 IAM role** scoped to AWS License Manager (to verify your Marketplace contract entitlement) + AWS Systems Manager (for keyless admin access).
- **1 security group** (UI access locked to a CIDR you choose).
- Optional: an **Elastic IP** and an **HTTPS Application Load Balancer** (with your ACM certificate).

Refine connects **read-only** to your AWS accounts, Azure subscriptions, and GCP projects and ranks cost-optimization opportunities across all three clouds in one queue.

## Prerequisites

- An AWS account with permission to launch CloudFormation (creates EC2, S3, IAM, SG).
- A VPC and a public subnet in your chosen region.
- The CIDR you'll allow to reach the Refine UI (your office/VPN egress IP, e.g. `203.0.113.10/32` — find yours at <https://checkip.amazonaws.com>).
- (Optional, for HTTPS) an ACM certificate in the same region + two subnets in different AZs for the load balancer.

## Step 1 — Subscribe on AWS Marketplace

1. Open the Blacktip Refine listing and choose **View purchase options**.
2. Choose the **contract tier** that matches your annual Spend Analyzed (Starter / Core / Growth / Scale) for a **1-year** term, and accept the terms (the EULA is the **Download EULA(s)** link).

3. Choose **Subscribe**. You'll land on the **Purchase confirmation** page once the agreement is **Active**. This contract entitlement is what Refine checks at launch — there is no separate license file to request.

## Step 2 — Launch the software

1. On the confirmation page, choose **Launch your software**.
2. On the Launch page, under **Service**, select **ECS** simply to proceed.

**Important:** Refine deploys as an EC2 host via the bundled CloudFormation template — **not** as an ECS/EKS task. Use the **CloudFormation deployment template** linked under **Deployment templates** (right side). Do not try to run the image directly on ECS/EKS.

3. Click the **CloudFormation deployment template** link. It opens the template in the CloudFormation console (Create stack).

## Step 3 — Configure the stack

On **Specify stack details**, set the parameters:

| Parameter                  | What to enter                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Stack name                 | e.g. refine                                                                                                     |
| VpcId                      | your VPC                                                                                                        |
| SubnetId                   | a <b>public</b> subnet in that VPC                                                                              |
| AllowedAppCidr             | the CIDR allowed to reach the UI (e.g. 203.0.113.10/32 )                                                        |
| EnableHttps                | <code>false</code> for a quick start (HTTP on port 8000); <code>true</code> if you have an ACM cert + 2 subnets |
| InstanceType               | <code>t3.medium</code> (default) is sufficient for most deployments                                             |
| AdminEmail / AdminPassword | the administrator login you want for the Refine UI — you'll sign in with these at first login                   |
| Everything else            | leave the defaults (Watchtower auto-update on, rollback monitor on, Marketplace Contract fulfillment)           |

Continue through **Configure stack options** (defaults are fine), then on **Review** check "**I acknowledge that AWS CloudFormation might create IAM resources with custom names**" and choose **Submit**.

## Step 4 — Wait for the stack, then read the Outputs

Wait ~8 minutes for **CREATE\_COMPLETE**. Open the **Outputs** tab and note:

- **RefineUrl1** — where the UI will be served (e.g. `http://<ip>:8000`).
- **AdminConnectMethod** — a one-click AWS Systems Manager Session Manager link for keyless shell access to the host (no SSH key needed).

The EC2 pulls the Refine image immediately and, on boot, automatically verifies your AWS Marketplace contract entitlement through AWS License Manager — no license file and no activation step.

## Step 5 — Automatic license activation

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There is nothing to do here. On boot, Refine calls **AWS License Manager** to verify the Marketplace contract entitlement tied to your subscription, records your tier, and the `refine` container comes up **healthy** on its own — usually within a minute of `CREATE_COMPLETE`. No license-request form, no emailed file, no `aws s3 cp`.

You can watch progress from the host via the `AdminConnectMethod` SSM link: `docker ps` (the `refine` container should go to `healthy`).

**Direct sales / GovCloud (BYOL):** if you obtained Refine outside AWS Marketplace — a direct purchase or an air-gapped GovCloud delivery — you instead receive a signed `refine.License` file plus an activation code from Blacktip and drop it into your delivery bucket, where an on-host watcher installs it. See the direct-sales deployment guide for that flow. The public Marketplace contract path above does **not** use it. (A Private Offer is a Marketplace contract, so it uses the entitlement path above — not BYOL.)

## Step 6 — First login

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Open `RefineUrl`. Log in with the **AdminEmail / AdminPassword** you set as stack parameters in Step 3. **Change your password immediately after first login** (Settings → Account).

## Step 7 — Connect your cloud accounts

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From **Accounts** → **Add Account**, connect each cloud (read-only):

- **AWS** — one-click CloudFormation onboarding stack (creates an IAM role + a Cost Explorer collector). Repeat per AWS account you want analyzed.
- **Azure** — a one-line `curl | bash` in Azure Cloud Shell (creates an Entra ID app registration, Reader role, and a Cost Management export). Auto-detects Commercial vs US Government.
- **GCP** — a Terraform module (service account + viewer roles + BigQuery billing export).

First recommendations and the cross-cloud savings ranking appear within minutes of the first sync.

## Keeping Refine up to date

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Auto-update is on by default: a Watchtower sidecar polls AWS Public ECR on a schedule (default daily 03:00 UTC) and updates the Refine container in place when a new version is published — with an automatic rollback to the last-known-good image if a new build fails its health check. To pin a version, set `EnableAutoUpdate=false` and a specific `RefineImageUri` tag.

## Troubleshooting

| Symptom                       | Check                                                                                                                                                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI won't load                 | Confirm your current IP matches <b>AllowedAppCidr</b> ; for HTTP, the URL is <code>http://&lt;ip&gt;:8000</code> .                                                                                                                      |
| "License" / entitlement error | Confirm the Marketplace subscription shows <b>Active</b> , and that the stack's IAM role can reach AWS License Manager ( <code>license-manager:CheckoutLicense</code> , <code>GetEntitlements</code> ). Refine re-checks automatically. |
| Container keeps restarting    | Expected briefly on first boot while Refine verifies the contract entitlement; it settles to <b>healthy</b> once the check succeeds.                                                                                                    |
| Need a shell on the host      | Use the <b>AdminConnectMethod</b> SSM Session Manager link from the stack Outputs — no SSH key required.                                                                                                                                |

## Support

- **Setup / licensing:** support@blacktip-ops.com
- **Sales / tiers:** sales@blacktip-ops.com